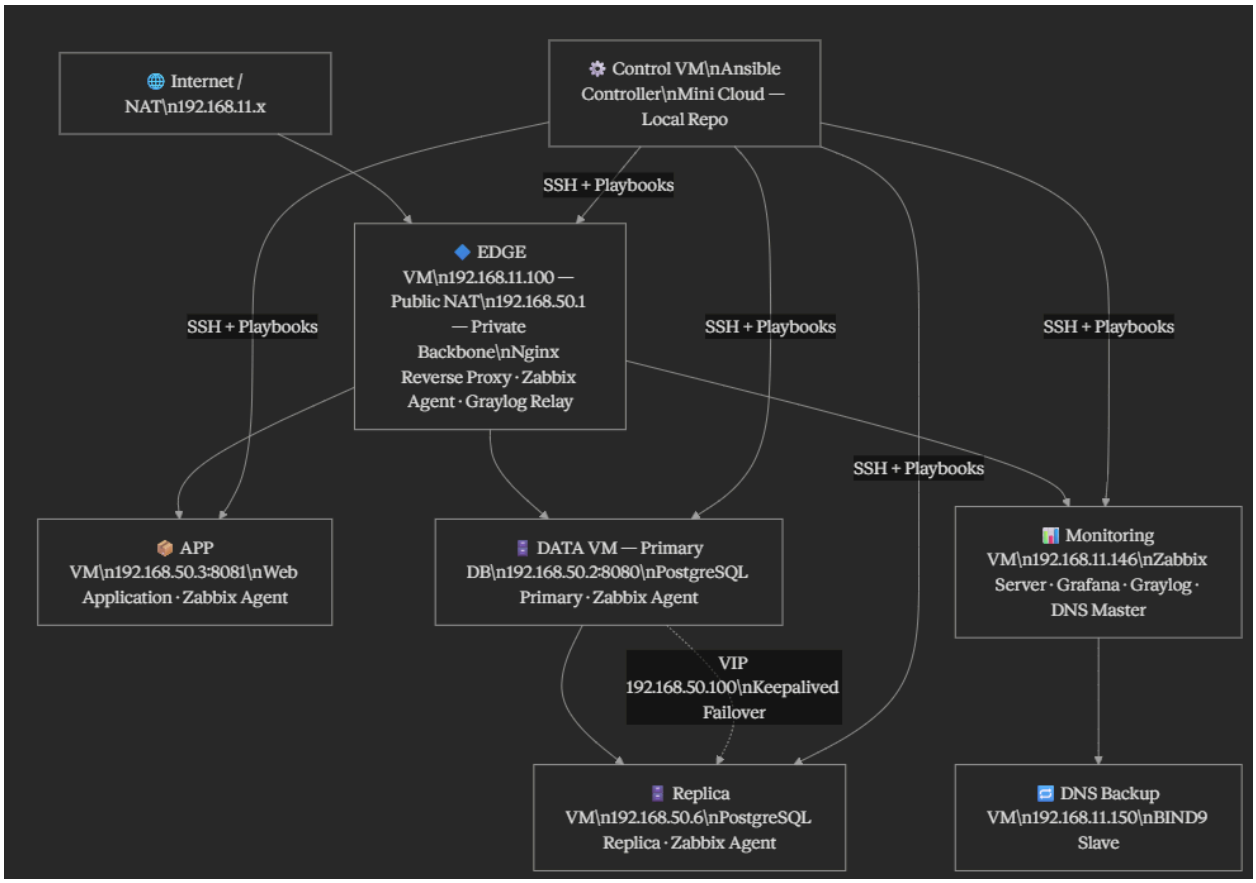
🖥️ Automated Infrastructure & Monitoring System

Personal Home Lab Project | RHEL 9 | 2025–2026

A production-grade home lab environment demonstrating real-world skills in infrastructure design, automation, high availability, and full-stack observability — built entirely from scratch on virtual machines.

🌐 **Project Website** | 📓 **Interactive Notebook (NotebookLM)** | 📄 **Full Report (PDF)**

📐 Architecture Overview



✅ Completed Milestones

Phase	Component	Status
Week 1	Internal VLAN + Reverse Proxy + VIP	✅
Week 1	Live System Health Dashboard	✅
Week 1	Nginx Access & Error Logging	✅
Week 2–3	Zabbix Monitoring + Agents on all VMs	✅
Week 2–3	PostgreSQL Primary/Replica Replication	✅
Week 2–3	Virtual IP — Keepalived HA Failover	✅
Week 2–3	Grafana Dashboard (CPU, Memory, Network)	✅
Week 2–3	Graylog Centralised Log Aggregation	✅
Week 3–4	BIND9 DNS Master/Slave (Forward + Reverse Zones)	✅
Week 3–4	Ansible Full Automation (5 Playbooks)	✅

🧱 Stack

Category	Technology
OS	RHEL 9
Containers	Podman + systemd services

Category	Technology
Web / Proxy	Nginx (containerized)
Database	PostgreSQL (Primary + Replica)
HA / Failover	Keepalived (Virtual IP)
Monitoring	Zabbix Server + Agents, Grafana
Logging	Graylog + OpenSearch + rsyslog
DNS	BIND9 (Master/Slave, Forward/Reverse zones)
Automation	Ansible (Playbooks + Jinja2 Templates)
Security	SELinux, firewalld, TLS-PSK (Zabbix)

🔧 Key Components

1. Nginx Reverse Proxy (Containerised)

- Routes `/app` → `192.168.50.3:8081` and `/data` → `192.168.50.2:8080`
- Custom `default.conf` with structured log format for monitoring
- Built into a custom image with embedded health-check scripts
- Managed as systemd services for reboot persistence

2. Live System Health Dashboard

- Bash script polls all backend services every 5 seconds via `curl`
- Auto-generates and updates `status.html` with service name, IP:port, HTTP status, and response time

3. PostgreSQL Primary/Replica Replication

- Primary has full read/write; Replica is read-only streaming replica
- Verified by inserting test data on Primary and reading instantly from Replica

4. Virtual IP — Keepalived HA

- VIP `192.168.50.100` floats between Primary and Replica DB VMs
- Health-check script uses `podman exec ... pg_isready` to test DB liveness

- On Primary failure, VIP migrates to the replica automatically — no manual steps

5. Zabbix + Grafana Monitoring

- Agents on all VMs report to the central Zabbix server
- Custom item: `systemd.unit.get["container-edge.service"]` + JSONPath `$.ActiveState.text` → returns `1` or `0`
- A single Grafana dashboard shows disk, CPU, memory, and network for all VMs
- Podman socket spoofed as Docker socket, so Zabbix agent2's built-in Docker plugin works with Podman

6. Graylog Centralised Logging

- Internal VMs forward logs → Edge VM (dual-homed) → Graylog container
- Ports: `1514/udp`, `1514/tcp`, `1515/tcp` to prevent protocol lock

7. BIND9 DNS Master/Slave

- Forward zone `monitoring.local` resolves all hosts on both `.11` and `.50` subnets
- Reverse PTR zones for both subnets
- Slave DNS auto-pulls all zone files from the Master
- Security: ACL, firewalld, SELinux `named_zone_t`, Ansible-managed patching

8. Ansible Automation

```
.
├── ansible.cfg
├── host_vars
│   └── web
│       └── main.yml
├── inventory
│   ├── group_vars
│   └── production
├── playbooks
│   ├── deploy_containers.yml
│   └── patch.yml
├── roles
│   ├── common
│   │   └── tasks
│   │       └── main.yml
│   ├── db
│   ├── dns-server
│   │   ├── handlers
│   │   │   └── main.yml
│   │   ├── tasks
│   │   │   └── main.yml
│   │   └── templates
│   │       ├── monitoring.local.forward
│   │       ├── monitoring.local.reverse.11
│   │       ├── monitoring.local.reverse.50
│   │       └── named.conf.j2
│   ├── firewall
│   │   └── tasks
│   │       └── main.yml
│   ├── graylog_forward
│   │   ├── handlers
│   │   │   └── main.yml
│   │   ├── tasks
│   │   │   └── main.yml
│   │   └── templates
│   │       └── graylog.conf.j2
│   ├── podman_stack
│   ├── web
│   │   ├── files
│   │   │   └── template.zip
│   │   ├── handlers
│   │   │   └── main.yml
│   │   └── tasks
│   │       └── main.yml
│   └── zabbix_agent
└── site.yml
```

A single `ansible-playbook site.yml` run configures Edge VM as NAT gateway, pushes DNS to all VMs, sets subnet-specific gateways, builds the Mini Cloud local package mirror, and deploys Graylog forwarding and web templates across the fleet.

🐛 Notable Challenges Solved

Problem	Root Cause	Solution
Containers dying on reboot	Podman is daemonless	`podman generate systemd --new` → systemctl service
`podman stop` conflicting with systemd	Two controllers owning same container	Only use `systemctl` — never raw `podman stop/start`
Log files created as symlinks	Wrong `COPY` path in Dockerfile	Fixed destination path in Dockerfile
Keepalived exit code 127	`podman` called without an absolute path	Use `/usr/bin/podman --url unix:///run/podman/podman.sock`
SELinux `exec_died` on VIP script	Missing SELinux context on script	`restorecon -v /usr/local/bin/check_pg.sh` + socket perms
VIP not migrating on failover	Missing `mcast_src_ip` in keepalived config	Added `mcast_src_ip` for the correct VRRP multicast source
Graylog connection refused	Container-bound UDP-only, input on TCP	Redeploy with both `/tcp` and `/udp` on both ports
Zabbix is showing inverted 0/1	Raw `ActiveState` is text, not numeric	JSONPath `$.ActiveState.text` + regex `active` → `1`

🔗 Resources

Resource	Link
🌐 Project Website	automatedinfra.lovable.app
📓 Interactive Notebook	NotebookLM
📄 Full Weekly Report	Project_Report.pdf

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